



DP-602: Implement a data warehouse with Microsoft Fabric

Exercise 2 : Load data into a warehouse using T-SQL

- Create a workspace
- Create a lakehouse and upload files
- Create a table in the lakehouse
- Create a warehouse
- Create fact table, dimensions and view
- Load data to the warehouse
- Run analytical queries



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Create a workspace

Create a workspace

Name *

Data_load_tsqI

✔

This name is available

Description

Describe this workspace

Domain ⓘ

Assign to a domain (optional)

Learn more about workspace settings

Workspace image

Upload

Reset

Advanced ^

Contact list * ⓘ

d(

dagstudentsgreatlearning (Owner)

×

Workspace type ⓘ

What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#)

Power BI only workspace types

Power BI workspaces only support the use of Power BI capabilities.

Apply

Cancel



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► Create a lakehouse and upload files

New Lakehouse

Name *

Location *

 MS Fabric Practice ▼ ↗

☒ Lakehouse schemas ⓘ

CreateCancel

tsql_LH

No label

Search

Trials activated: 28 days left

1

Home

Materialized lake views

Get data

New semantic model

Add to data agent

Manage OneLake security

Update all variables

For optimal refresh to work efficiently, enable the delta CDF property on both source tables and materialized lake views used as source.

A SQL analytics endpoint for SQL querying was created with this item.

Explorer

tsql_LH

Tables

dbo

Files

Files

MS Fabric Practice > tsql_LH > Files

Showing 1 items

Name	Date modified	Type	Size
salesj.csv	6/16/2026, 4:03:53 ...	csv	3 MB



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► Create a table in the lakehouse

Load file to new table

All fields marked with * are required

Schema *

dbo

New table name *

staging_sales

Column header ⓘ



Use header for column names

Separator ⓘ

,

Separators cannot use the following characters: () [] { } ' "

Load

Cancel



DP-602: Implement a data warehouse with Microsoft Fabric

► Create a warehouse

New Warehouse

Name *

Location *

 MS Fabric Practice ▼ ↗

Advanced

Create

Cancel

Explorer

+ Warehouses

▼ T-SqlWH

> Schemas

> Security

▼ Queries

> My queries

> Shared queries

Build a warehouse

Start getting data



New Copy job

Keep warehouse tables in sync with scheduled data transfers



New Dataflow Gen2

Ingest and transform data with a low-code interface.



Practice with sample data

Automatically load sample data in your warehouse.

Start developing



T-SQL

Create tables with T-SQL code templates.



SQL database project

Publish a database schema from an external project.

[Need help? Learn more](#)

 /Julurishashi



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➤ Create fact table, dimensions and view

Home Management Help Analyze data with

Explorer

- Warehouses
 - INFORMATION_SCHEMA
 - queryinsights
 - Sales**
 - Tables
 - Dim_Customer
 - Dim_Item
 - Fact_Sales
 - Views
 - Functions
 - Stored Procedures
 - sys
 - Security
- Queries
 - My queries
 - SQL query 1**
 - Shared queries

SQL query 1

Run Save as Save as view Create rule Explain query Fix query errors

```
1 CREATE SCHEMA [Sales]
2 GO
3
4 IF OBJECT_ID('Sales.Fact_Sales', 'U') IS NULL
5     CREATE TABLE Sales.Fact_Sales (
6         CustomerID VARCHAR(255) NOT NULL,
7         ItemID VARCHAR(255) NOT NULL,
8         SalesOrderNumber VARCHAR(30),
9         SalesOrderLineNumber INT,
10        OrderDate DATE,
11        Quantity INT,
12        TaxAmount FLOAT,
13        UnitPrice FLOAT
14    );
15
16 IF OBJECT_ID('Sales.Dim_Customer', 'U') IS NULL
17     CREATE TABLE Sales.Dim_Customer (
18         CustomerID VARCHAR(255) NOT NULL,
19         CustomerName VARCHAR(255) NOT NULL,
20         EmailAddress VARCHAR(255) NOT NULL
21    );
22
23 ALTER TABLE Sales.Dim_Customer add CONSTRAINT PK_Dim_Customer PRIMARY KEY NONCLUSTERED (CustomerID) NOT ENFORCED
24 GO
25
26 IF OBJECT_ID('Sales.Dim_Item', 'U') IS NULL
27     CREATE TABLE Sales.Dim_Item (
```

Messages Copy messages



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► Load data to the warehouse

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 - My queries
 - SQL query 1
 - SQL query 2

SQL query 2

```
1 CREATE OR ALTER PROCEDURE Sales.LoadDataFromStaging (@OrderYear INT)
2 AS
3 BEGIN
4     -- Load data into the Customer dimension table
5     INSERT INTO Sales.Dim_Customer (CustomerID, CustomerName, EmailAddress)
6     SELECT DISTINCT CustomerName, CustomerName, EmailAddress
7     FROM [Sales].[Staging_Sales]
8     WHERE YEAR(OrderDate) = @OrderYear
9     AND NOT EXISTS (
10         SELECT 1
11         FROM Sales.Dim_Customer
12         WHERE Sales.Dim_Customer.CustomerName = Sales.Staging_Sales.CustomerName
13         AND Sales.Dim_Customer.EmailAddress = Sales.Staging_Sales.EmailAddress
14     );
15
16     -- Load data into the Item dimension table
17     INSERT INTO Sales.Dim_Item (ItemID, ItemName)
18     SELECT DISTINCT Item, Item
19     FROM [Sales].[Staging_Sales]
20     WHERE YEAR(OrderDate) = @OrderYear
21     AND NOT EXISTS (
22         SELECT 1
23         FROM Sales.Dim_Item
```

Messages

4:29:03 PM Started executing on line 1

4:29:03 PM Query execution time: 00:00:00.017 | Time to process and render results: 00:00:00.796 | Total duration: 00:00:00.813



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► Run analytical queries

SQL query 1 SQL query 2 SQL query 3

Run Save as Save as view Create rule Explain query Fix query

```
1 SELECT c.CustomerName, SUM(s.UnitPrice * s.Quantity) AS TotalSales
2 FROM Sales.Fact_Sales s
3 JOIN Sales.Dim_Customer c
4 ON s.CustomerID = c.CustomerID
5 WHERE YEAR(s.OrderDate) = 2021
6 GROUP BY c.CustomerName
7 ORDER BY TotalSales DESC;
```




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► Run analytical queries

```
SQL query 1 | SQL query 2 | SQL query 3 | SQL query 4 X
```

```
Run | Save as | Save as view | Create rule | Explain query | Fix query errors
```

```
1 WITH CategorizedSales AS (  
2 SELECT  
3     CASE  
4         WHEN i.ItemName LIKE '%Helmet%' THEN 'Helmet'  
5         WHEN i.ItemName LIKE '%Bike%' THEN 'Bike'  
6         WHEN i.ItemName LIKE '%Gloves%' THEN 'Gloves'  
7         ELSE 'Other'  
8     END AS Category,  
9     c.CustomerName,  
10    s.UnitPrice * s.Quantity AS Sales  
11 FROM Sales.Fact_Sales s  
12 JOIN Sales.Dim_Customer c  
13 ON s.CustomerID = c.CustomerID  
14 JOIN Sales.Dim_Item i  
15 ON s.ItemID = i.ItemID  
16 WHERE YEAR(s.OrderDate) = 2021  
17 ),  
18 RankedSales AS (  
19     SELECT  
20         Category,  
21         CustomerName,  
22         SUM(Sales) AS TotalSales,  
23         ROW_NUMBER() OVER (PARTITION BY Category ORDER BY SUM(Sales) DESC) AS SalesRank  
24     FROM CategorizedSales  
25     WHERE Category IN ('Helmet', 'Bike', 'Gloves')  
26     GROUP BY Category, CustomerName  
27 )
```



DP-602: Implement a data warehouse with Microsoft Fabric

► Reference

